ERIOLUWA AYODEJI

PETROLEUM ENGINEER & ENERGY AI SYSTEMS ARCHITECT

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EXECUTIVE PROFILE

Petroleum Engineering graduate (B.Eng., Second-Class Upper) from Covenant University and technical founder pioneering real-time physics-informed AI systems for upstream drilling dynamics, predictive asset maintenance, and regulatory digital twins. Founder & Systems Architect of the **PetroOne Energy Intelligence Suite**, including **Omesham AI** (autonomous drilling telemetry & vibration dampening) and **GAIA.AI**. Combines deep subsurface domain knowledge with advanced algorithmic engineering across high-frequency SCADA/WITSML streaming, Aspen HYSYS process simulation, and industrial safety design.

TECHNICAL PROFICIENCIES & DOMAIN MASTERY

Drilling & Petroleum Engineering: Downhole Telemetry (WITSML/LAS), Torsional Stick-Slip Mitigation, Rate of Penetration (ROP) & Weight on Bit (WOB) Optimization, BHA Directional Trajectory, Well Intervention, Aspen HYSYS, HAZOP/HAZID, PFDs/P&IDs.;

AI, Systems & Architecture: Physics-Informed Neural Networks, Python (NumPy, SciPy, Pandas), Real-time Time-Series Analytics, Next.js/React, Micro-frontend SSO Handshakes, Serverless Cloud Architecture, REST/WebSocket APIs, PostgreSQL.

FLAGSHIP ENERGY & AI CREATIONS

Omesham AI — Autonomous Physics-Informed Drilling Co-Pilot — Lead Architect

- Engineered an upstream telemetry co-pilot ingesting high-frequency downhole data (WITSML/LAS) across active geothermal and oil basins (Utah FORGE, Permian Basin).
- **Torsional Vibration Mitigation:** Modeled drillstring wave equations to detect and suppress torsional stick-slip resonance, mitigating destructive lateral vibrations (>3G) and drillstring twist-offs.
- **Autonomous State Classification:** Built real-time classifiers distinguishing Sliding vs. Rotating vs. Tripping states with live directional BHA toolface azimuth drift tracking.
- **Predictive Hydraulics:** Developed transient Standpipe Pressure (SPP) models detecting downhole mud-motor micro-stalls and BHA washouts up to 30 minutes before mechanical manifestation.
- Pitched at executive level to independent operators including **First E&P; (Ademola Adeyemi-Bero)** for offshore Niger Delta shallow-water assets.

PetroOne Energy Intelligence Suite (PetroSight, GAIA.AI, Petrogenesis 3D) — Full-Stack Founder

- Architected a unified energy micro-frontend suite with orbital single sign-on (SSO) bridging **Omesham AI**, **PetroSight AI** (predictive maintenance & asset health), **GAIA.AI** (upstream lead and field intelligence), and **Petrogenesis 3D** (subsurface borehole visualization).
- Engineered fault-tolerant telemetry stream handlers managing live operational alarms with zero-latency visual status reporting.

NUPRC Digital Twin & Regulatory Safety Framework — Technical Author

- Designed an enterprise digital twin architecture for regulatory oversight with the **Nigerian Upstream Petroleum Regulatory Commission (NUPRC)**, focusing on real-time hydrocarbon accounting and offshore safety integrity.

Mindflow AI — Voice-First Executive Intelligence Engine — Creator

- Deployed a voice-to-structured-data AI assistant using Groq Whisper (<0.5s audio transcription) and multimodal LLMs for automated task, date, and fiscal expense structuring deployed serverless on Vercel.

PROFESSIONAL INDUSTRY EXPERIENCE

National Engineering & Technical Company (NETCO - NNPC Subsidiary) — Mar 2024 – Sept 2024

Engineering Intern — Process & Upstream Facility Engineering

- Performed process simulations in Aspen HYSYS for real-world oil and gas upstream production and gas treatment facilities.
- Contributed to commercial Hazard Identification (HAZID) and Hazard and Operability (HAZOP) studies for facility debottlenecking.
- Developed practical exposure to well intervention procedures, including coiled tubing operations, cementing, and high-pressure nitrogen purging.

Tomesy Engineering Nigeria Limited — Jul 2023 – Aug 2023

Engineering Intern — Process Plant Systems

- Drafted industrial Process Flow Diagrams (PFDs) and assisted with engineering risk analysis conforming to HSE standards.

EDUCATION & PROFESSIONAL AFFILIATIONS

Covenant University, Ota, Nigeria — *B.Eng., Petroleum Engineering* **2020 – 2025**

- **Academic Distinction:** Graduated Second-Class Honours (Upper Division).
- **Leadership:** Assistant Membership Lead, Society of Petroleum Engineers (SPE), Covenant University (2024 – 2025).
- **Certifications:** Aspen HYSYS Process Simulation, Data Analysis with Python (DataCamp), HSE Level 1 & 2.